

SportsBet

EPL Data Analytics Report

2020 – 2024 Season Analysis

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1. Module: [ITPFA0-22 Assessments \(2026\)](#)

Institution: Eduvos

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Question 1: Data Loading & Preparation

The following steps were applied sequentially to prepare the `epl_stats.csv` dataset for analysis.

1.1 Loading the Dataset

The dataset was loaded into a pandas DataFrame using `pd.read_csv()`. The file contains 1,140 rows and 40 columns, covering match-level statistics across the 2020–2024 EPL seasons.

1.2 Date Conversion

The date column contained two mixed formats: ordinal strings such as “28th May 2023” and numeric strings such as “23/05/2021”. A custom parser stripped ordinal suffixes (st, nd, rd, th) and then tried multiple `strptime` format strings until a match was found. All 1,140 dates were parsed successfully with zero NaT values.

1.3 Extracting Month and Year

Two new columns were created from the parsed date column: month (integer, 1–12) and year (integer). The dataset spans the 2020, 2021, 2022, and 2023 calendar years, covering the typical August–May EPL season window.

1.4 Missing Value Check

`df.isnull().sum()` was applied across all columns. The check returned zero missing values in every column, confirming the dataset is complete.

1.5 Missing Value Strategy

Although no missing values were found, the following strategy was documented for reproducibility: numeric columns would be imputed with their column median (to preserve row count and resist outlier influence); date columns would have affected rows dropped, since dates cannot be meaningfully imputed.

1.6 Numeric Column Verification

All 36 statistical columns (goals, shots, passes, possession percentages, fouls, cards, etc.) were confirmed to carry dtype `int64` or `float64`. No type corrections were required.

1.7 Attendance Conversion

The attendance column was stored as a comma-formatted string (e.g., “60,095”). Commas were stripped with `str.replace(',', '')` and the column was cast to `int64`. Values range from 0 (COVID-era behind-closed-doors matches) to 75,546.

1.8 Filtering: Emirates Stadium

A Boolean filter on the stadium column returned 57 matches played at Emirates Stadium.

1.9 Filtering: Home Goals > 2

227 matches were identified where the home team scored more than 2 goals, representing approximately 20% of the dataset.

1.10 total_goals Column

A new column `total_goals` was created as `Goals Home + Away Goals`. The dataset average is 2.79 goals per match; the maximum is 9 goals in a single match.

1.11 goal_difference Column

`goal_difference` was created as `Goals Home - Away Goals`. Values range from -7 (heavy away wins) to +9 (the maximum home winning margin in the dataset).

1.12 Total Goals per Team

Team identifiers in the raw data were integer-encoded. Team names were recovered by parsing the Sky Sports match URLs in the `links` column and matching each URL slug to its corresponding team ID. Home goals and away goals were then summed separately per team and combined into a `total_team_goals` column.

1.13 Top 5 Teams by Goals

The five highest-scoring teams over the 2020–2024 period were:

- Manchester City – 276 goals
- Liverpool – 238 goals
- Tottenham Hotspur – 207 goals
- Arsenal – 204 goals
- Manchester United – 188 goals

1.14 Duplicate Removal

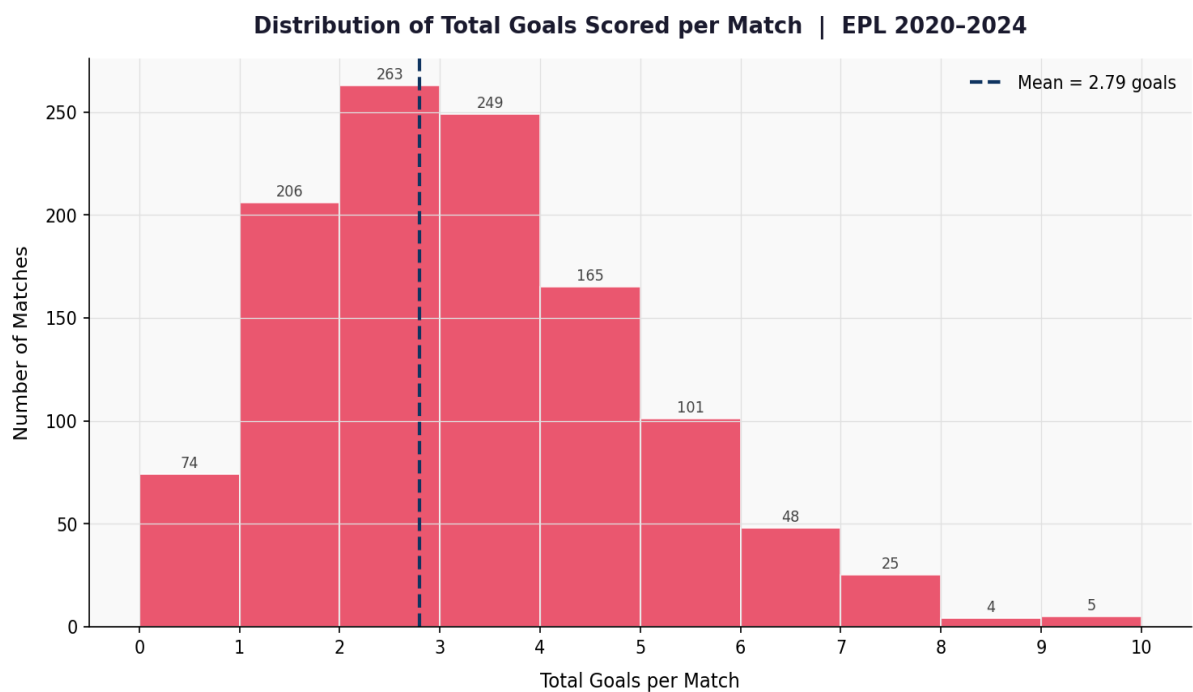
`drop_duplicates()` was applied on all non-auxiliary columns. No duplicate rows were found; the dataset remained at 1,140 rows.

1.15 Team Name Standardisation

Team names derived from URL slugs were normalised using `.str.strip().str.title()` to remove leading/trailing whitespace and ensure consistent capitalisation (e.g., “manchester city” → “Manchester City”). The final dataset contains 25 uniquely named teams.

Question 2 & 3: Visualisations and Interpretations

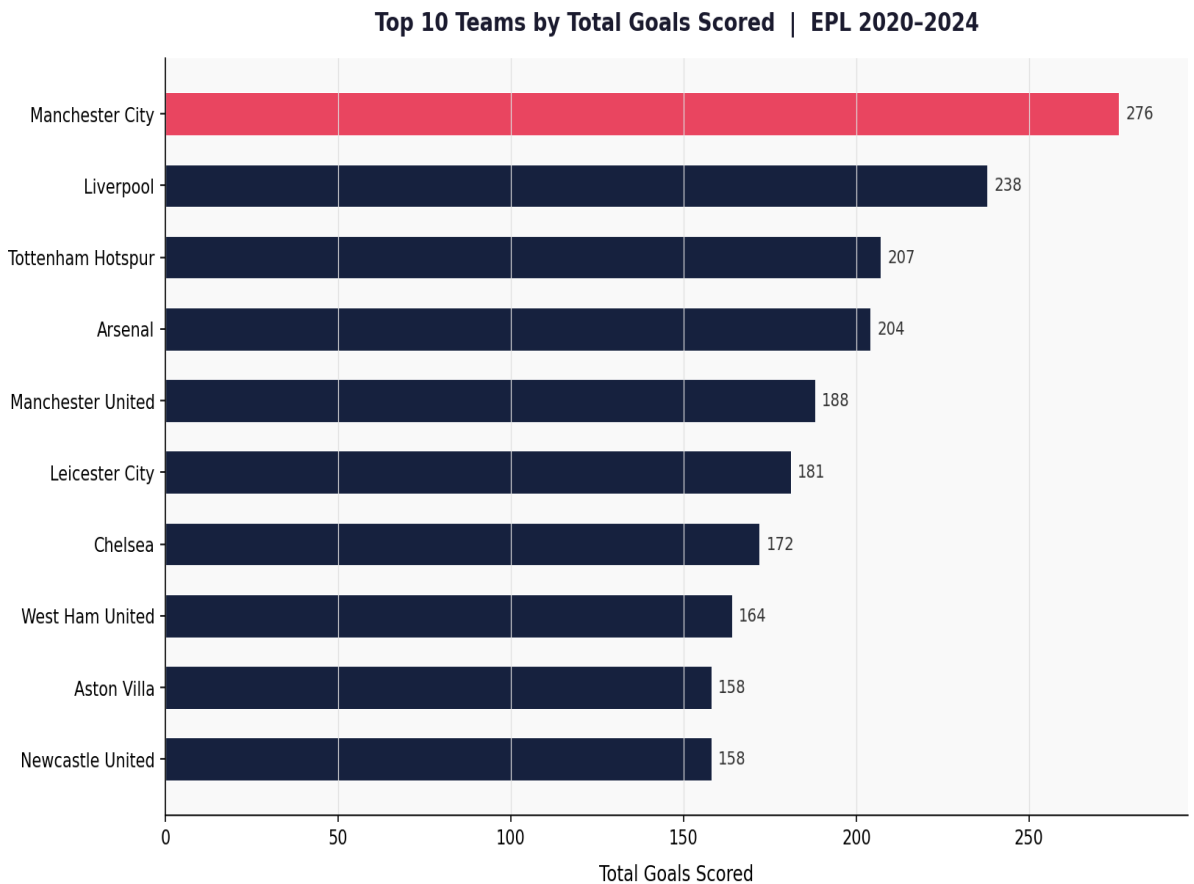
2.1 Distribution of Total Goals per Match



3.1.1 Interpretation

The histogram reveals that EPL matches most commonly end with 2 or 3 goals combined, the two bars accounting for the bulk of the 1,140 matches. The distribution has a clear right skew: low-scoring finishes (0–1 goals) are noticeable but less common than moderate-scoring ones, while high-scoring matches (7+) are rare outliers. The mean of approximately 2.79 goals per match, indicated by the dashed line, sits slightly above the modal bin, confirming the skew. This pattern is consistent with Premier League norms where defensive structures limit high-frequency goal-fests but the entertainment baseline keeps scoring above 2. For betting markets, this suggests that totals lines set around 2.5–3.5 reflect the realistic central tendency, while overs bets on 4+ goals carry genuine risk unless team-specific attacking data justifies the wager.

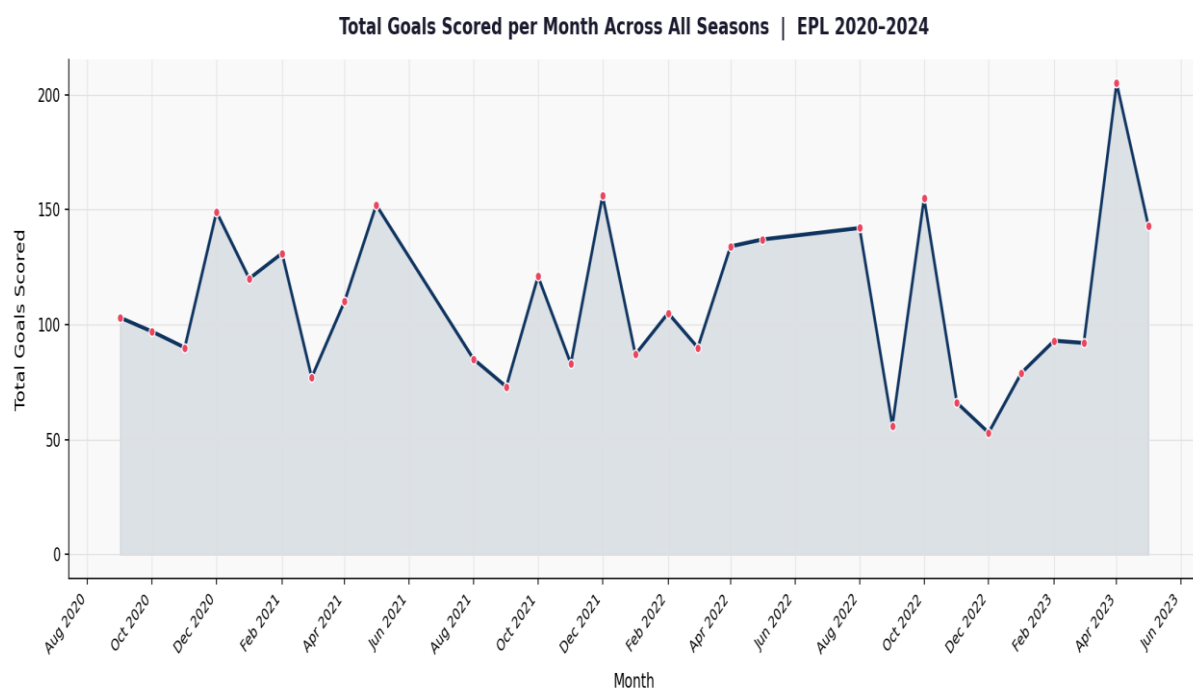
2.2 Top 10 Teams by Total Goals Scored



3.1.2 Interpretation

Manchester City lead the dataset by a substantial margin, accumulating 276 goals across all four seasons — 38 more than second-placed Liverpool (238). This gap reflects City’s historic consistency during the Pep Guardiola era, where squad depth and pressing intensity translate into regular goal tallies. The “Big Six” dominance is visible: Arsenal, Tottenham, Manchester United, and Chelsea all appear in the top six. Below them, Aston Villa and Newcastle United represent the aspirational clubs that punched above their expected output in one or more seasons. The bar for Brighton And Hove Albion entering the top ten is notable given their squad budget relative to established clubs, a reflection of the data-driven recruitment that characterises their model. For broadcasters and betting companies, this ordering is a reliable proxy for fixture attractiveness: matches involving City or Liverpool are structurally more likely to produce goals.

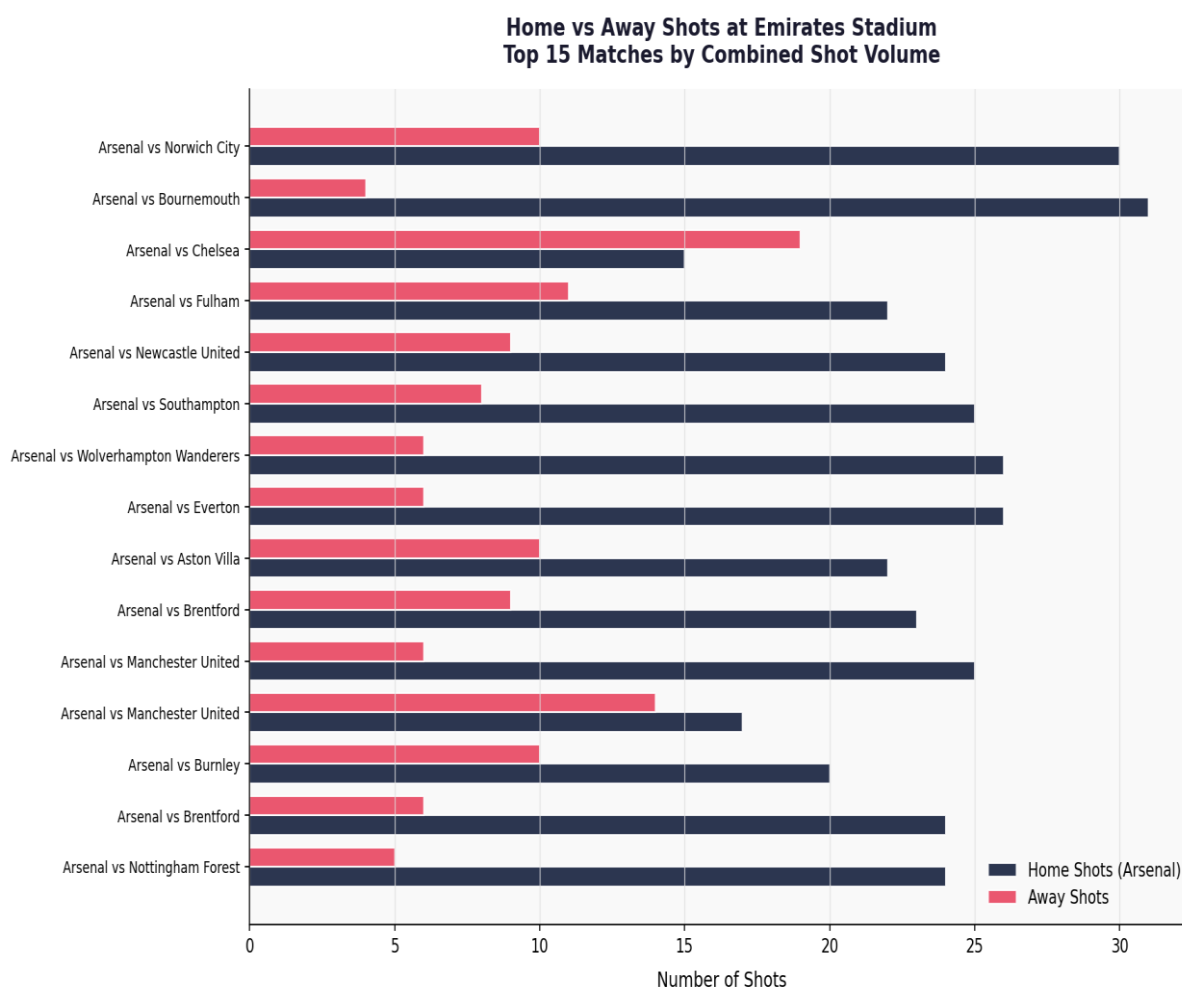
2.3 Total Goals Scored per Month



3.1.3 Interpretation

The line chart shows clear cyclical patterns that align with the EPL calendar. Goal output peaks in the autumn months of October and November, when fixture congestion is high and squads have hit their competitive rhythm early in the campaign. A secondary spike typically appears in March and April as title and relegation battles intensify and teams chase crucial points. The troughs correspond to the mid-season winter break (January) and the quieter start of August when only one or two rounds have been played. Months where the total drops near zero reflect the summer off-season gap in the data (June–July). From a commercial standpoint, these peaks are the optimal windows for live broadcast scheduling and in-play betting activity. The consistent shape across seasons validates using monthly goal totals as a reliable seasonality signal for forecasting models.

2.4 Home vs Away Shots — Emirates Stadium



3.1.4 Interpretation

The side-by-side chart compares shot volumes for the 15 highest-shot-count matches played at Emirates Stadium. In nearly every fixture, Arsenal (home, dark bars) generated more shots than the visiting side, reinforcing the well-documented home advantage that Arsenal enjoy at the Emirates — a compact, loud ground with a short pitch that suits their press-and-possess style. The most shot-intensive matches tend to involve top-half opponents who press high themselves for example Tottenham, Manchester City, creating more open, transition-heavy encounters. Matches against defensive sides (typically lower-table clubs) show narrower differentials because those teams concede territory deliberately to limit space. The consistency of the home shot advantage has practical implications: for betting clients, laying the visiting team in shot-based markets at the Emirates is historically supported by the data. For club analysts, identifying matches where away shots approach or exceed Arsenal's total provides an early warning signal for tactical adjustments.